THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The nucleus is absent in mature: 1

(a) muscle cells **(b)** liver cells
(c) red blood cells of mammals **(d)** nerve cells

2. A beetroot piece placed in hot water loses its colour into the water because: 1

(a) the cell membrane is damaged **(b)** the nucleus bursts
(c) the wall dissolves **(d)** osmosis occurs

3. Which of the following is the smallest unit capable of independent life? 1

(a) cell **(b)** organelle **(c)** tissue **(d)** molecule

4. A tissue culture can produce: 1

(a) no plants **(b)** many identical plants from a small piece
(c) only seeds **(d)** new species

5. A cell without a cell wall and chloroplast but with centrioles is a: 1

(a) fungal cell **(b)** plant cell **(c)** bacterium **(d)** animal cell

6. Wilted leaves become fresh in water because: 1

(a) nucleus divides **(b)** cells absorb water by osmosis
(c) cells lose water **(d)** stomata close

7. The longest cell in the human body is: 1

(a) muscle cell **(b)** sperm **(c)** RBC **(d)** nerve cell

8. Stem cells are important because they: 1

(a) cannot divide
(b) can develop into different types of cells
(c) are dead
(d) are only in plants

9. The genetic material in bacteria lies in the: 1

(a) nucleus **(b)** plastid **(c)** mitochondrion **(d)** nucleoid

10. Salting of pickles preserves them because: 1

(a) bacteria lose water by osmosis **(b)** bacteria gain water
(c) salt cooks them **(d)** salt gives vitamins

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are cells small in size? Give two reasons. 2

12. State why a cell with more mitochondria would be found in muscle tissue. 2

13. Why does a freshwater Amoeba have a contractile vacuole? 2

14. State why chloroplasts and mitochondria are called semi-autonomous organelles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the cell wall and the cell membrane (three points). 3

16. State the importance of cell division in single-celled and multicellular organisms. 3

17. A cell was placed in a solution of 0.9 per cent salt and shows no change. State the type of solution and the reasoning. 3

OR

Explain why wilted flowers revive in water. Use the terms turgor and osmosis. 3

18. A cell has a large Golgi apparatus. Suggest the function of such a cell and give an example. 3

OR

Explain how the cell organelles work together to produce and secrete insulin. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the experiment to demonstrate plasmolysis using Rhoeo or onion peel, with observations and conclusion. 5

OR

Explain the importance of osmosis in plants (absorption, turgidity, opening of stomata, wilting) and in daily life (pickles). 5

20. Explain the process and stages of mitosis with simple descriptions and why it is important for growth and repair. 5

OR

Describe the differences in the structure and function of different animal cells (RBC, nerve, muscle, sperm, white blood cell) and relate to the number of organelles. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND